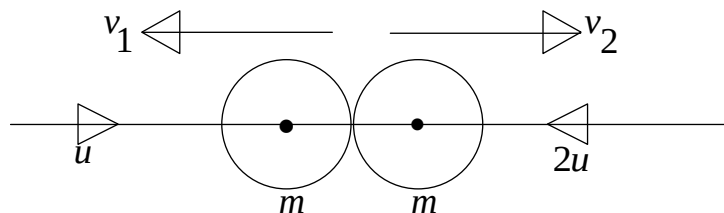


Combined Maths Applied

**Revision – 2026 – Test – 10
Model Marking Scheme**

Janaka Abeywardana

01.



නිව්ටන්ගේ ප්‍රත්‍යාගති නියමයෙන් ,

$$v_1 + v_2 = e(u + 2u) \quad (05)$$

$$v_1 + v_2 = 3eu \quad \text{-----} (1)$$

—→ ගම්‍යතා සංස්ථිති නියමයෙන් ,

$$mv_2 - mv_1 = -2mu + mu \quad (05)$$

$$v_2 - v_1 = -u \quad \text{-----} (2)$$

$$(1) + (2) \text{ න් } \Rightarrow 2v_2 = (3e - 1)u$$

$$v_2 = (3e - 1)\frac{u}{2}$$

$$(1) - (2) \text{ න් } \Rightarrow 2v_1 = (3e + 1)u$$

$$v_1 = (3e + 1)\frac{u}{2}$$

$$\begin{aligned} v_1^2 + v_2^2 &= \left[(3e + 1)\frac{u}{2} \right]^2 + \left[(3e - 1)\frac{u}{2} \right]^2 \\ &= 2[9e^2 + 1]\frac{u^2}{2 \times 2} \end{aligned}$$

$$\therefore v_1^2 + v_2^2 = [9e^2 + 1]\frac{u^2}{2} \quad (05)$$

ගැටුමෙන් පසු චාලක ශක්තිය = $\frac{1}{2}$ (ගැටුමට පෙර චාලක ශක්තිය)

$$\frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}\left(\frac{1}{2}mu^2 + \frac{1}{2}m(2u)^2\right) \quad (05)$$

$$\div \frac{m}{2} \Rightarrow v_1^2 + v_2^2 = \frac{5u^2}{2}$$

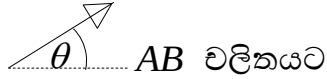
$$(9e^2 + 1)\frac{u^2}{2} = \frac{5u^2}{2}$$

$$9e^2 = 4$$

$$e = \pm \frac{2}{3}$$

$$e \geq 0 \text{ නිසා } e = \frac{2}{3} \text{ වේ. } (05)$$

02.



$$v = u + at$$

$$0 = \sqrt{2ga} + (-g \sin \theta) t \quad (05)$$

$$t = \frac{\sqrt{2ga}}{g \sin \theta} \quad (05)$$

AB වලිකයට

$$\left. \begin{aligned} v \sin \theta &= \sqrt{2ga} \cos \theta \\ -v \cos \theta &= \sqrt{2ga} \sin \theta - gt \end{aligned} \right\} (05)$$

$$gt = \sqrt{2ga} \sin \theta + \sqrt{2ga} \cot \theta \cos \theta$$

$$t = \sqrt{\frac{2a}{g}} \left(\frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta} \right) \Rightarrow t = \frac{\sqrt{2ga}}{g \sin \theta} \quad (05)$$

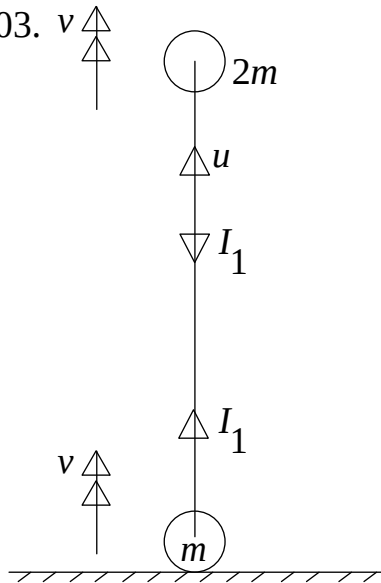
$$AB \text{ වලිකයට} \rightarrow S = ut + \frac{1}{2}at^2$$

$$R = \sqrt{2ga} \cos \theta \cdot \frac{\sqrt{2ga}}{g \sin \theta}$$

$$R = 2a \cot \theta \quad (05)$$

25

03.



B හි මුල් වලිකයට

$$\uparrow v^2 = u^2 + 2aS \text{ යෙදීමෙන්,}$$

$$u^2 = 4ga + 2(-g)a$$

$$u = \sqrt{2ga} \quad (05)$$

$$A \text{ ට } \uparrow I = \Delta(mv) \text{ යෙදීමෙන්,}$$

$$I_1 = mv \text{ ----- (1) } (05)$$

$$B \text{ ට } \uparrow I = \Delta(mv) \text{ යෙදීමෙන්,}$$

$$-I_1 = 2mv - 2m\sqrt{2ga} \text{ --(2) } (05)$$

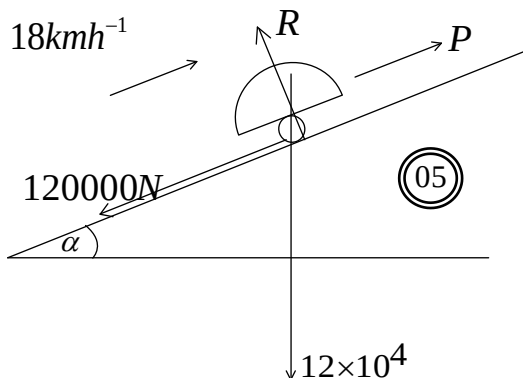
$$(1) + (2) \Rightarrow 0 = 3mv - 2m\sqrt{2ga}$$

$$v = \frac{2\sqrt{2ga}}{3} \quad (05)$$

$$(1) \text{ ට } \Rightarrow I_1 = \frac{2m\sqrt{2ga}}{3} \quad (05)$$

25

04.



$$\alpha = \frac{\pi}{6}$$

$$12.96 \text{ kmh}^{-2} = \frac{12.96 \times 10^3}{(3600)^2} \text{ ms}^{-2}$$

$$= \frac{1.296}{36 \times 36} = \frac{1}{1000} \text{ ms}^{-2}$$

$$18 \text{ kmh}^{-2} = \frac{18 \times 10^3}{3600} \text{ ms}^{-2}$$

$$= 5 \text{ ms}^{-2}$$

$$\alpha \quad \underline{F} = m \underline{a}$$

$$P - 120000 - 12 \times 10^3 \times 10 \sin 30^\circ = 12 \times 10^3 \times \frac{1}{1000} \quad (10)$$

$$P = 120000 + 12 \times 10^4 \times \frac{1}{2} + 12$$

$$P = 120000 + 60000 + 12$$

$$= \underline{\underline{180012 \text{ N}}}$$

$$H = PV$$

$$= 180012 \times 5 \quad (05)$$

$$= 900060 \text{ W}$$

$$\underline{\underline{H = 900.06 \text{ kW}}} \quad (05)$$



05. $\uparrow R \cos \alpha - mg = 0 \quad (05)$

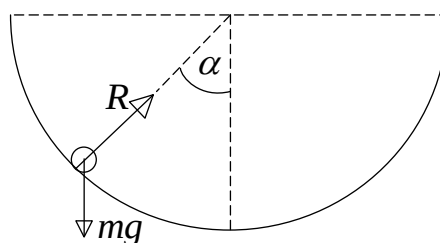
$$R = \frac{mg}{\cos \alpha}$$

$$\rightarrow F = ma$$

$$R \sin \alpha = \frac{mu^2}{a \sin \alpha} \quad (05)$$

$$\frac{mg}{\cos \alpha} \cdot \sin^2 \alpha = \frac{mu^2}{a}$$

$$\underline{\underline{\frac{\sin^2 \alpha}{\cos \alpha} = \frac{u^2}{ga}}} \quad (05)$$



$$\alpha = \frac{\pi}{3} \text{ වීම } \Rightarrow \frac{\sin^2 \frac{\pi}{3}}{\cos \frac{\pi}{3}} = \frac{u^2}{ga}$$

$$u^2 = \frac{ga \left(\frac{\sqrt{3}}{2} \right)^2}{\left(\frac{1}{2} \right)} = \frac{3ga}{2}$$

$$u = \sqrt{\frac{3ga}{2}} \quad (05)$$

$$a \sin \frac{\pi}{3} \omega = \sqrt{\frac{3ga}{2}} ; \quad \because v = r\omega \text{ අනුව}$$

$$a \times \frac{\sqrt{3}}{2} \cdot \frac{2\pi}{T} = \sqrt{\frac{3ga}{2}} ; \quad \because \omega = \frac{2\pi}{T}$$

$$T = \pi \sqrt{\frac{2a}{g}} \quad (05)$$



06. $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$

$$\overrightarrow{AC} = \lambda \overrightarrow{AB}$$

$$\overrightarrow{AB} = -\underline{a} + \underline{b}$$

$$\overrightarrow{AC} = \lambda(-\underline{a} + \underline{b})$$

$$\begin{aligned} \overrightarrow{OC} &= \overrightarrow{OA} + \overrightarrow{AC} \\ &= \underline{a} + \lambda(-\underline{a} + \underline{b}) \end{aligned}$$

$$\overrightarrow{OC} = (1 - \lambda)\underline{a} + \lambda\underline{b} \quad (05)$$

$$\overrightarrow{OC} \perp \overrightarrow{AB} \text{ නම් ,}$$

$$\overrightarrow{OC} \cdot \overrightarrow{AB} = 0 \text{ විය යුතු යි.}$$

$$((1 - \lambda)\underline{a} + \lambda\underline{b}) \cdot (-\underline{a} + \underline{b}) = 0 \quad (05)$$

$$-(1 - \lambda)\underline{a} \cdot \underline{a} + \lambda\underline{b} \cdot \underline{b} + (1 - \lambda)\underline{a} \cdot \underline{b} - \lambda\underline{a} \cdot \underline{b} = 0$$

$$\lambda|\underline{b}|^2 - (1 - \lambda)|\underline{a}|^2 = (2\lambda - 1)\underline{a} \cdot \underline{b}$$

$$\underline{a} \cdot \underline{b} = \frac{\lambda|\underline{b}|^2 - (1 - \lambda)|\underline{a}|^2}{2\lambda - 1} \text{ ----- (1) } (05)$$

$$\hat{AOB} = \frac{\pi}{2} \text{ නිසා}$$

$$\overrightarrow{OA} \perp \overrightarrow{OB} \text{ වේ. } \therefore \underline{a} \cdot \underline{b} = 0 \text{ වේ. } (05)$$

$$(1) \text{ න් } \Rightarrow 0 = \frac{\lambda|\underline{b}|^2 - (1 - \lambda)|\underline{a}|^2}{2\lambda - 1}$$

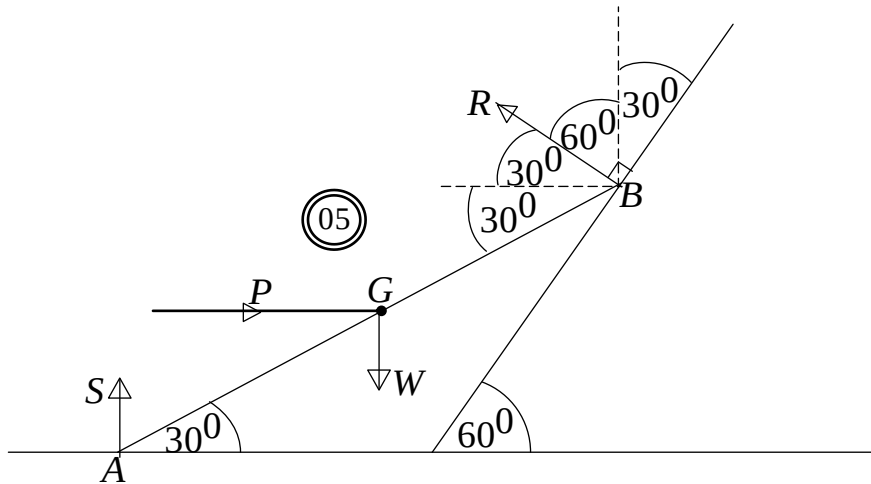
$$\lambda|\underline{b}|^2 = (1 - \lambda)|\underline{a}|^2$$

$$\left(\frac{|\underline{b}|}{|\underline{a}|} \right)^2 = \frac{1-\lambda}{\lambda}$$

$$\frac{|\underline{b}|}{|\underline{a}|} = \sqrt{\frac{1-\lambda}{\lambda}} \quad (05) \quad \left(\because \frac{|\underline{b}|}{|\underline{a}|} > 0 \right)$$



07.



$$P - R \cos 30^\circ = 0 \quad (05)$$

$$P - \frac{\sqrt{3}}{2} R = 0$$

$$R = \frac{2}{\sqrt{3}} P \quad (05)$$

$$AB = 2l \quad (05)$$

$$\curvearrowleft A \quad R \times 2l \cos 30^\circ - P l \sin 30^\circ - W l \cos 30^\circ = 0 \quad (05)$$

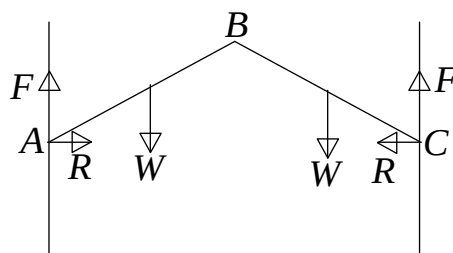
$$\frac{2P}{\sqrt{3}} \times 2 \times \frac{\sqrt{3}}{2} - P \times \frac{1}{2} - W \frac{\sqrt{3}}{2} = 0$$

$$2P - \frac{P}{2} = \frac{\sqrt{3}W}{2}$$

$$= \frac{W}{\sqrt{3}} \quad (05)$$



08.



$$\uparrow 2F - 2W = 0 \quad (05)$$

$$F = W$$

දෘශ්‍යය දිග $2a$ යැයි සිතමු.

BC ට $\triangle B$ වටා

$$F \times 2a \sin 60^\circ - R \times 2a \cos 60^\circ - W \times a \sin 60^\circ = 0 \quad (05)$$

$$W \times 2 \times \frac{\sqrt{3}}{2} - W \times \frac{\sqrt{3}}{2} = R \times 2 \times \frac{1}{2}$$

$$\underline{\underline{R = \frac{\sqrt{3}W}{2}}} \quad (05)$$

සමතුලිතව පැවතීමට ,

$$\left. \begin{array}{l} \frac{F}{R} \leq \mu \\ \frac{W}{\frac{\sqrt{3}W}{2}} \leq \mu \end{array} \right\} (05)$$

$$\mu \geq \frac{2}{\sqrt{3}}$$

$$\underline{\underline{\mu \geq \frac{2\sqrt{3}}{3}}} \quad (05)$$

25

$$09. \quad P(A' \cap B') = P((A \cup B)') = 1 - P(A \cup B) \quad (05)$$

$$P(A \cup B) = 1 - P(A' \cap B')$$

$$P(A \cup B) = 1 - \frac{3}{8} = \frac{5}{8} \quad (05)$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \quad (05)$$

$$P(A \cap B) = \frac{1}{3} + \frac{1}{2} - \frac{5}{8} = \frac{5}{24} \quad (05)$$

$$P(A \cap B') = P(A) - P(A \cap B)$$

$$= \frac{1}{3} - \frac{5}{24}$$

$$= \frac{1}{8} \quad (05)$$

25

10. නිව්ල 5 = {x, y, y, 3y, 3y}

$$\text{මධ්‍යන්‍යය} = \frac{x + y + y + 3y + 3y}{5} = 2x \text{ (05)}$$

$$8y = 9x$$

$$x = 8 \text{ විට } 8y = 9x$$

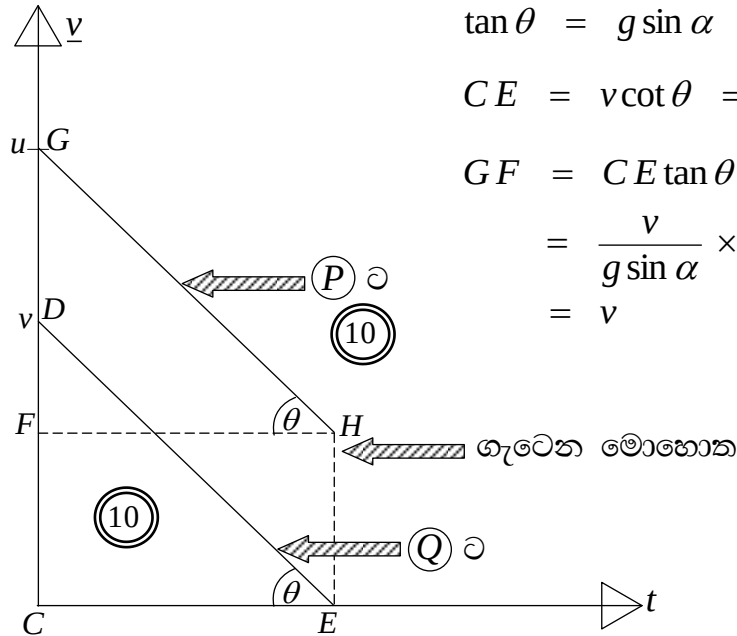
$$8y = 9 \times 8 \Rightarrow \underline{y = 9} \text{ (05)}$$

$$\text{නිව්ල 5} = \{8, 9, 9, 27, 27\} \text{ (05)}$$

$$\left. \begin{array}{l} a = \text{Min} = 8 \\ e = \text{Max} = 27 \\ c = \bar{x} = 9 \end{array} \right\} \text{ (05)}$$

$$\left. \begin{array}{l} b = Q_1 = \frac{8+9}{2} = 8.5 \\ d = Q_3 = \frac{27+27}{2} = 27 \end{array} \right\} \text{ (05)}$$

11. (a)



$$\tan \theta = g \sin \alpha$$

$$CE = v \cot \theta = \frac{v}{g \sin \alpha}$$

$$\begin{aligned} GF &= CE \tan \theta \\ &= \frac{v}{g \sin \alpha} \times g \sin \alpha \\ &= v \end{aligned}$$

ගැටෙන මොහොත

20

$$\left. \begin{aligned} Q \text{ වලින දුර } + h &= P \text{ වලින දුර} \\ CDE \text{ වර්ගඵලය } + h &= CGHE \text{ වර්ගඵලය} \\ \text{දෙපසින්ම } CDE \text{ වර්ගඵලය අඩු කල විට,} \\ h &= GDEH \text{ වර්ගඵලය} \\ h &= (u - v)CE \\ &= (u - v) \frac{v}{g \sin \alpha} \end{aligned} \right\} \textcircled{05}$$

$$gh \sin \alpha = uv - v^2$$

$$v^2 - uv + gh \sin \alpha = 0 \quad \textcircled{05}$$

$$v = \frac{u \pm \sqrt{u^2 - 4gh \sin \alpha}}{2} \quad \textcircled{05}$$

$$v < \frac{u}{2} \text{ නිසා}$$

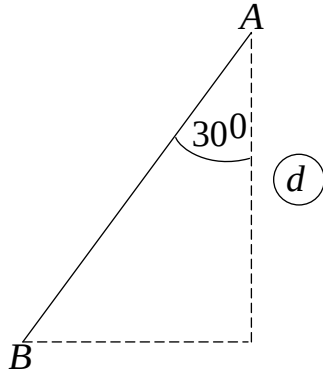
$$v = \frac{u - \sqrt{u^2 - 4gh \sin \alpha}}{2} \quad \textcircled{05}$$

$$\left. \begin{aligned} P \text{ අංශුව } Q \text{ හි ගැටෙන ප්‍රවේගය} &= u - GF \\ &= \underline{\underline{u - v}} \end{aligned} \right\} \textcircled{10}$$

40

60

(b)



$$\underline{V}(B, E) = \underline{u}$$

$$\underline{V}(A, E) = v$$

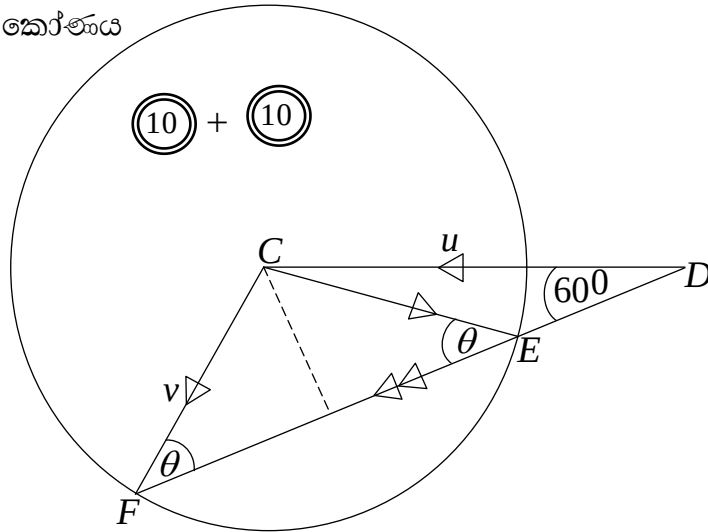
$$\begin{aligned} AB &= d \sec 30^\circ \\ &= \underline{\underline{\frac{2d}{\sqrt{3}}}} \end{aligned}$$

සාපේක්ෂ ප්‍රවේග මූලධර්මයෙන් ,

$$\underline{V}(A, B) = \underline{V}(A, E) + \underline{V}(E, B)$$

$$30^\circ = v + u \quad (15)$$

ප්‍රවේග ත්‍රිකෝණය

එකම චලිතය සඳහා CDE ත්‍රිකෝණය සහ CDF ත්‍රිකෝණය ලෙස ප්‍රවේග ත්‍රිකෝණ දෙකක් (05)ඇඳිය හැකි නිසා , $A \cap B$ හමුවියහැකි සේ චලිත විය හැකි දිශා දෙකක් පවතී.

$$\sin \theta = \frac{u \sin 60^\circ}{v} \quad (05)$$

$$= \frac{\sqrt{3}u}{2v}$$

$$\theta = \sin^{-1} \frac{\sqrt{3}u}{2v} \quad (05)$$

$$\text{එම දිශා දෙක අතර කෝණය} = \pi - 2\theta \quad (05)$$

$$= \pi - 2 \sin^{-1} \frac{\sqrt{3}u}{2v} \quad (05)$$



$$\underline{V}(A, B) = \frac{u}{2} + \frac{\sqrt{4v^2 - 3u^2}}{2} \quad (05) \text{ කෝ}$$

$$\underline{V}(A, B) = \frac{u}{2} - \frac{\sqrt{4v^2 - 3u^2}}{2} \quad (05)$$

$$t = \frac{AB \sin \theta}{|\underline{V}(A, B)|}$$

$$\begin{aligned} t_1 &= \frac{2d/\sqrt{3}}{\left\{ \frac{u + \sqrt{4v^2 - 3u^2}}{2} \right\}} \\ &= \frac{4d}{\sqrt{3}(u + \sqrt{4v^2 - 3u^2})} \quad (05) \quad (1) \end{aligned}$$

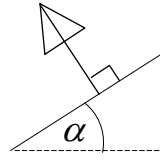
$$\begin{aligned} t_2 &= \frac{2d/\sqrt{3}}{\left\{ \frac{u - \sqrt{4v^2 - 3u^2}}{2} \right\}} \\ &= \frac{4d}{\sqrt{3}(u - \sqrt{4v^2 - 3u^2})} \quad (05) \quad (2) \end{aligned}$$

$$\begin{aligned} \text{කාල අන්තරය} &= t_2 - t_1 \\ &= \frac{4d}{\sqrt{3}(u - \sqrt{4v^2 - 3u^2})} - \frac{4d}{\sqrt{3}(u + \sqrt{4v^2 - 3u^2})} \quad (05) \\ &= \frac{4d}{\sqrt{3}} \left(\frac{2\sqrt{4v^2 - 3u^2}}{u^2 + 3u^2 - 4v^2} \right) \\ &= \frac{2d\sqrt{4v^2 - 3u^2}}{\sqrt{3}(u^2 - v^2)} \\ &= \left[\frac{4d^2(4v^2 - 3u^2)}{3(u^2 - v^2)^2} \right]^{1/2} \quad (05) \end{aligned}$$

150

30

90

පද්ධතියටම  $F = m a$

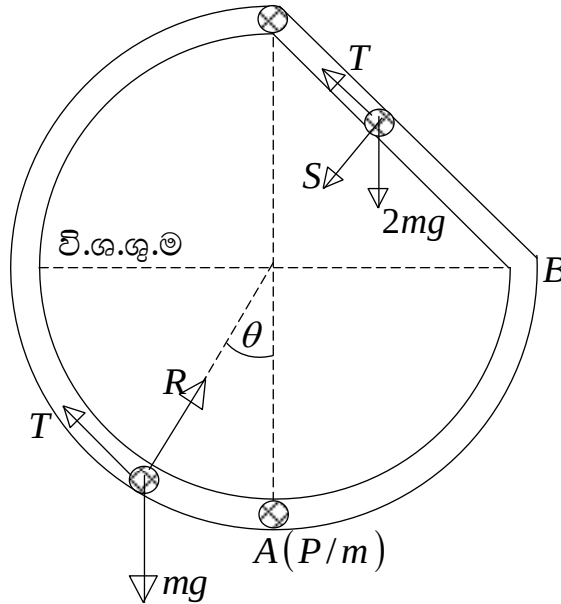
$$S - Mg \cos \alpha - mg \cos \alpha - mg \cos \alpha \quad (20)$$

$$= M(0) + m(f \sin \alpha) + m(-f \cos \alpha)$$

65

85

(b)



ශක්ති සංස්ථිති නියමයෙන් ,

$$\frac{1}{2} m (a \dot{\theta})^2 + \frac{1}{2} 2m (a \dot{\theta})^2 - mg a \cos \theta \quad (15)$$

$$+ 2mg \left(a - a \theta \cos \frac{\pi}{4} \right) = 2mg a - mg a$$

$$\div ma \Rightarrow$$

$$\frac{1}{2} \{ a \dot{\theta}^2 + 2 a \dot{\theta}^2 \} = g - 2g + \sqrt{2} g \theta + g \cos \theta \quad (05)$$

$$\underline{\underline{a \dot{\theta}^2 = \frac{2g}{3} [\sqrt{2} \theta - 1 + \cos \theta]}} \quad (05)$$

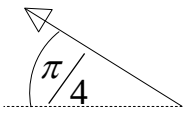
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 t විෂයෙන් අවකලනය ,

$$a 2 \dot{\theta} \ddot{\theta} = \frac{2g}{3} (\sqrt{2} \dot{\theta} + (-\sin \theta) \dot{\theta}) \quad (05)$$

$$\div 2 \dot{\theta} \Rightarrow$$

$$a \ddot{\theta} = \frac{g}{3} (\sqrt{2} - \sin \theta) \quad (05)$$



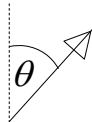
$$2m \odot \quad \underline{F} = m \underline{a}$$

$$T - 2mg \cos \frac{\pi}{4} = -2ma\ddot{\theta} \quad (05)$$

$$T = \sqrt{2}mg - 2m \left[\frac{g}{3} (\sqrt{2} - \sin \theta) \right]$$

$$T = \frac{mg}{3} (\sqrt{2} + 2\sin \theta) \quad (05)$$





$$\underline{F} = m \underline{a}$$

$$R - mg \cos \theta = ma\dot{\theta}^2 \quad (05)$$

$$R = mg \cos \theta + m \left[\frac{2g}{3} (\sqrt{2}\theta - 1 + \cos \theta) \right]$$

$$= mg \left[\cos \theta + \frac{2\sqrt{2}\theta}{3} - \frac{2}{3} + \frac{2\cos \theta}{3} \right] \quad (05)$$

$$\theta = \pi/4 \quad \ominus \odot$$

$$R = mg \left[\frac{1}{\sqrt{2}} + \frac{2\sqrt{2}(\pi/4)}{3} - \frac{2}{3} + \frac{2}{3\sqrt{2}} \right] \quad (05)$$

$$= mg \left[\frac{5}{3\sqrt{2}} - \frac{2}{3} + \frac{\pi}{3\sqrt{2}} \right]$$

$$R = \frac{mg}{3\sqrt{2}} (5 - 2\sqrt{2} + \pi) \quad (05)$$

150



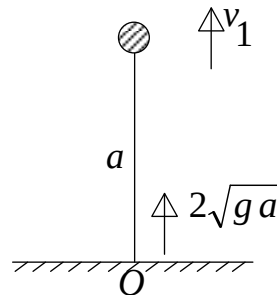
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11. $\uparrow v^2 = u^2 + 2a \cdot S$

$$v_1^2 = (2\sqrt{ga})^2 + 2(-g)a \quad (10)$$

$$v_1^2 = 4ga - 2ga = 2ga$$

$$v_1 > 0 \quad \text{සිසා} \quad \underline{v_1 = \sqrt{2ga}} \quad (05)$$



$$T = \frac{\lambda x}{\ell} \text{ ආනුච්ඡිද්‍රව}$$

$$T_1 = \frac{mgx}{2a}$$

$$\uparrow F = ma$$

$$-mg - T_1 = m\ddot{x}$$

$$-mg - \frac{mgx}{2a} = m\ddot{x} \quad (15)$$

$$\div m \Rightarrow \ddot{x} = -\frac{gx}{2a} - g = -\frac{g}{2a}(x + 2a)$$

$$\ddot{x} + \frac{g}{2a}(x + 2a) = 0 \quad (05)$$

$$\ddot{x} + \omega^2(x + 2a) = 0$$

$X = x + 2a$ නම් ,

t විෂයයෙන් වරක් අවකලනය

$$\dot{X} = \dot{x} \quad (05)$$

$$\therefore \ddot{X} = \ddot{x} \quad (05)$$

$$\therefore \ddot{X} + \omega^2 X = 0 \quad (05)$$



$x = 0$ විට $\dot{x} = \sqrt{2ga}$ වේ.

$$\dot{X}^2 = \omega^2(A^2 - X^2)$$

$$\dot{X}^2 = \omega^2[A^2 - (x + 2a)^2]$$

$$(\sqrt{2ga})^2 = \frac{g}{2a}[A^2 - (0 + 2a)^2] \quad (10)$$

$$4a^2 = A^2 - 4a^2$$

$$A^2 = 8a^2$$

$$A > 0 \text{ නිසා ; } A = 2\sqrt{2}a \quad (05)$$

කේන්ද්‍රයේ දී $\ddot{x} = 0$ නිසා

$$x + 2a = 0$$

$$x = -2a \quad (05)$$

$$\text{අංශුව 0 සිට ඉහළ නගින උපරිම උස} = 2\sqrt{2}a - a \quad (05)$$

$$= (2\sqrt{2} - 1)a \quad (05)$$



$$X = 2\sqrt{2}a \text{ විට } \dot{X} = -u \text{ වේ.}$$

$$\dot{X}^2 = \omega^2 (B^2 - X^2) \text{ ට අනුව,}$$

$$u^2 = \frac{g}{2a} \left[B^2 - (2\sqrt{2}a)^2 \right] \quad (05)$$

$$B^2 = \frac{2u^2a}{g} + 8a^2$$

$$B > 0 \text{ නිසා ; } B = \sqrt{\frac{2u^2a}{g} + 8a^2} \quad (05)$$

$$\text{තත්කූල බුරුල් වන විට } X = 2a$$

$$\dot{X}^2 = \omega^2 (B^2 - X^2) \text{ ට අනුව,}$$

$$(2\sqrt{ga})^2 = \frac{g}{2a} \left[\frac{2u^2a}{g} + 8a^2 - (2a)^2 \right] \quad (05)$$

$$\frac{2u^2a}{g} = 4a^2$$

$$u = \sqrt{2ga} \quad (05)$$

$$\text{එවිට } B = \sqrt{\frac{2(2ga)a}{g} + 8a^2} \quad (05)$$

$$= \sqrt{12a^2}$$

$$B = 2\sqrt{3}a \quad (05)$$

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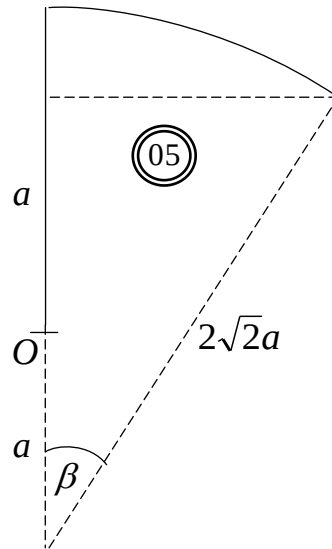
$$\cos \beta = \frac{2a}{2\sqrt{2}a}$$

$$= \frac{1}{\sqrt{2}} \quad (05)$$

$$\beta = \frac{\pi}{4}$$

$$\text{කාලය } (t_1) = \frac{\beta}{\omega}$$

$$= \sqrt{\frac{2a}{g}} \left(\frac{\pi}{4} \right) \quad (05)$$



$$\cos \alpha = \frac{2\sqrt{2}a}{2\sqrt{3}a} \quad (05)$$

$$= \frac{\sqrt{2}}{\sqrt{3}}$$

$$\alpha = \cos^{-1} \sqrt{\frac{2}{3}}$$

$$\cos \gamma = \frac{2a}{2\sqrt{3}a} \quad (05) = \frac{1}{\sqrt{3}}$$

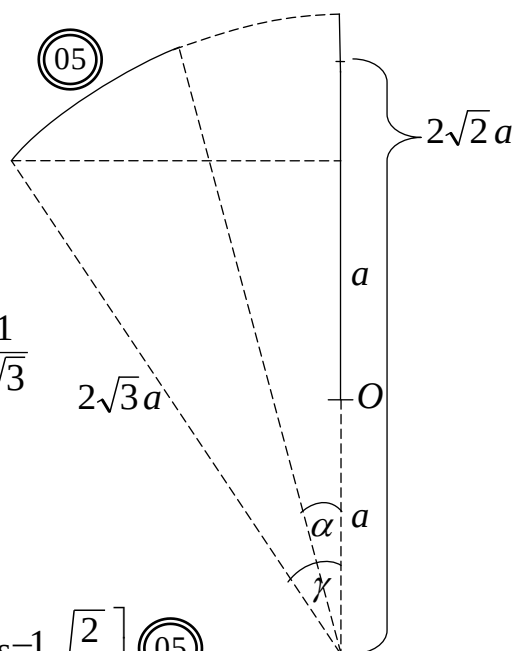
$$\gamma = \cos^{-1} \frac{1}{\sqrt{3}}$$

$$\text{කාලය}(t_2) = \frac{\gamma - \alpha}{\omega}$$

$$= \sqrt{\frac{2a}{g}} \left[\cos^{-1} \frac{1}{\sqrt{3}} - \cos^{-1} \sqrt{\frac{2}{3}} \right] \quad (05)$$

$$\therefore \text{මුළු කාලය} = t_1 + t_2$$

$$= \sqrt{\frac{2a}{g}} \left[\frac{\pi}{4} + \cos^{-1} \frac{1}{\sqrt{3}} - \cos^{-1} \sqrt{\frac{2}{3}} \right] \quad (05)$$

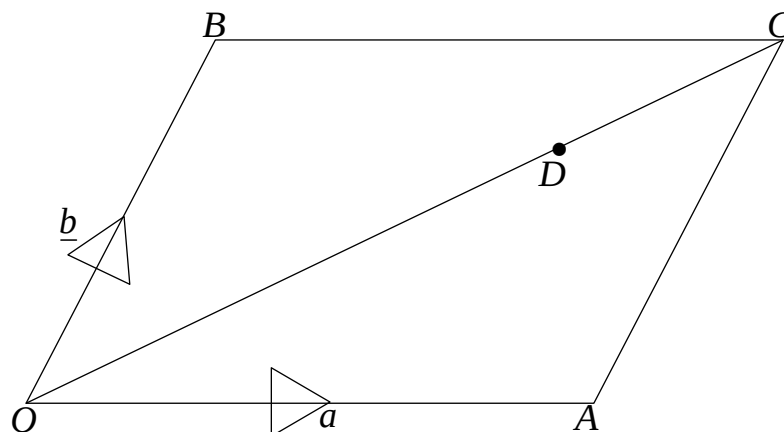


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14. (a)



$$\vec{AC} = \vec{OB} = \underline{b} \text{ වේ.}$$

$$\vec{OC} = \vec{OA} + \vec{AC}$$

$$= \underline{a} + \underline{b}$$

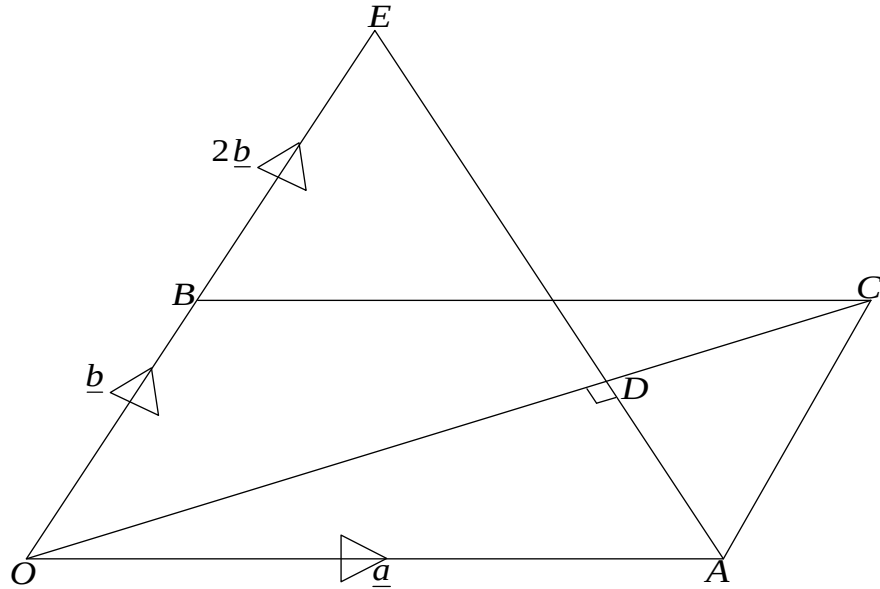
$$\vec{AD} = \vec{AO} + \vec{OD} \quad (05)$$

$$= -\underline{a} + \lambda(\vec{OC}) \quad (10)$$

$$= -\underline{a} + \lambda(\underline{a} + \underline{b})$$

$$= (\lambda - 1)\underline{a} + \lambda \underline{b}$$

15



$$\left. \begin{aligned} \overrightarrow{AD} &= \alpha \overrightarrow{AE} \\ (\lambda - 1)\underline{a} + \lambda \underline{b} &= \alpha (\overrightarrow{AO} + \overrightarrow{OE}) \\ (\lambda - 1)\underline{a} + \lambda \underline{b} &= \alpha (-\underline{a} + 2\underline{b}) \\ (\lambda + \alpha - 1)\underline{a} + (\lambda - 2\alpha)\underline{b} &= 0 \end{aligned} \right\} \textcircled{10}$$

$\underline{a} \neq 0$, $\underline{b} \neq 0$, $\underline{a} \times \underline{b} \neq 0$ නිසා ,

$$\underline{b} \Rightarrow \lambda - 2\alpha = 0 \quad \textcircled{05}$$

$$\lambda = 2\alpha$$

$$\underline{a} \Rightarrow \lambda + \alpha - 1 = 0 \quad \textcircled{05}$$

$$2\alpha + \alpha - 1 = 0$$

$$\alpha = \frac{1}{3} \Rightarrow \lambda = \frac{2}{3} \quad \textcircled{05}$$



$$\overrightarrow{AD} \perp \overrightarrow{OC} \text{ නිසා } \overrightarrow{AD} \cdot \overrightarrow{OC} = 0$$

$$\frac{1}{3}(2\underline{b} - \underline{a})(\underline{a} + \underline{b}) = 0 \quad \textcircled{10}$$

$$2\underline{b} \cdot \underline{b} + \underline{a} \cdot \underline{b} - \underline{a} \cdot \underline{a} = 0 \quad \textcircled{05}$$

$$2|\underline{b}||\underline{b}|\cos 0 + |\underline{a}||\underline{b}|\cos \theta - |\underline{a}||\underline{a}|\cos(0) = 0 \quad \textcircled{05}$$

$$|\underline{b}||\sqrt{2}|\underline{b}|\cos \theta = 2|\underline{b}|^2 - 2|\underline{b}|^2 \quad \textcircled{05}$$

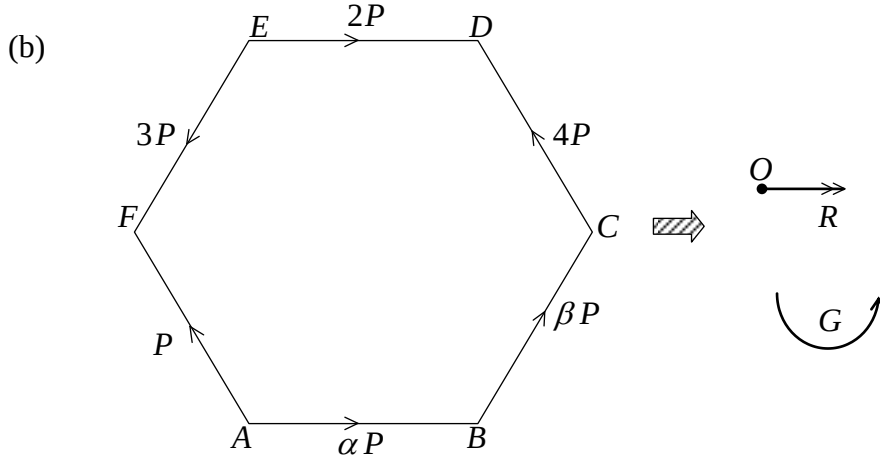
$$\cos \theta = 0$$

$$\therefore \theta = \pi/2 \text{ වේ. } \quad \textcircled{05}$$

; θ යනු $\underline{a}$ හා $\underline{b}$ අතර කෝණයයි.

OA හා OB එකිනෙකට ලම්භක වේ.





↑ බල විභේදනයෙන් ;

$$P \sin 60^0 + \beta P \sin 60^0 + 4P \sin 60^0 - 3P \sin 60^0 = 0 \quad (05)$$

$$1 + \beta + 4 - 3 = 0$$

$$\boxed{\beta = -2} \quad (05)$$

→ බල විභේදනයෙන් ;

$$\alpha P + \beta P \cos 60^0 + 2P - 4P \cos 60^0 \quad (05)$$

$$- 3P \cos 60^0 - P \cos 60^0 = R$$

$$R = \left[\alpha P - \frac{10P}{2} + 2P \right]$$

$$= (\alpha P - 3P)$$

$$= (\alpha - 3)P$$

$$|R| = |\alpha - 3|P \quad (05)$$



මගේ චරිත ;

$$\alpha P \cdot a \sin 60^0 + \beta P \cdot a \sin 60^0 + 4P \cdot a \sin 60^0 \quad (10)$$

$$+ (-2P \cdot a \sin 60^0) + 3P \cdot a \sin 60^0 - P \cdot a \sin 60^0 = G$$

$$G = 2Pa \sin 60^0 + \alpha P a \sin 60^0$$

$$= (\alpha + 2)Pa \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{3}(\alpha + 2)Pa}{2} \quad (05)$$

$$|R| = 5P$$

$$|\alpha - 3|P = 5P$$

$$\alpha - 3 = \pm 5 \quad (05)$$

$$\alpha = 8 \text{ හෝ } \alpha = -2$$

$$(05)$$

$$(05)$$



DC $\curvearrowright$ (D)

$$y(2\ell) - W\ell = 0 \quad (10)$$

$$2y = W$$

$$\underline{\underline{y = \frac{W}{2}}} \quad \text{-----} \quad (1) \quad (05)$$

BC $\curvearrowright$ (B)

$$W\ell \sin 30^\circ + y 2\ell \sin 30^\circ - x 2\ell \cos 30^\circ = 0 \quad (15)$$

$$\div \ell \Rightarrow$$

$$W \times \frac{1}{2} + 2y \times \frac{1}{2} - 2x \times \frac{\sqrt{3}}{2} = 0 \quad (05)$$

$$\sqrt{3}x - y = \frac{W}{2} \quad \text{-----} \quad (2)$$

$$(1) \text{ \& } (2) \Rightarrow$$

$$\sqrt{3}x - \frac{W}{2} = \frac{W}{2} \quad (05)$$

$$\sqrt{3}x = W$$

$$\underline{\underline{x = \frac{W}{\sqrt{3}}}} \quad (05)$$

50

AD, DC $\curvearrowright$ (A)

$$W\ell \sin 30^\circ + y\ell + x 2\ell \cos 30^\circ - T 2\ell \sin 30^\circ = 0 \quad (15)$$

$$\frac{W\ell}{2} + \frac{W}{2} \times \ell + 2x\ell \times \frac{\sqrt{3}}{2} - 2T\ell \times \frac{1}{2} = 0 \quad (05)$$

$$\div \ell \Rightarrow$$

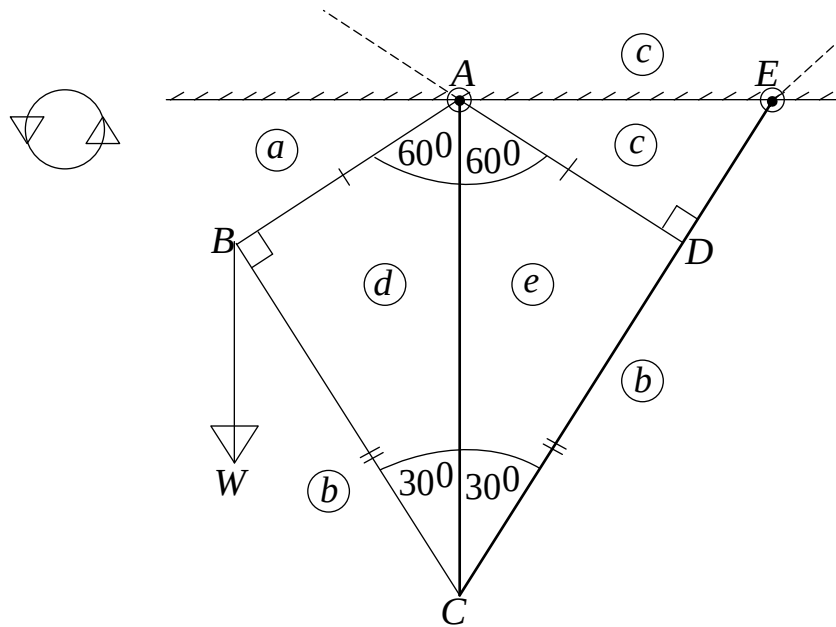
$$\frac{W}{2} + \frac{W}{2} + \frac{W}{\sqrt{3}} \times \sqrt{3} = T$$

$$(05) \quad \underline{\underline{T = 2W}}$$

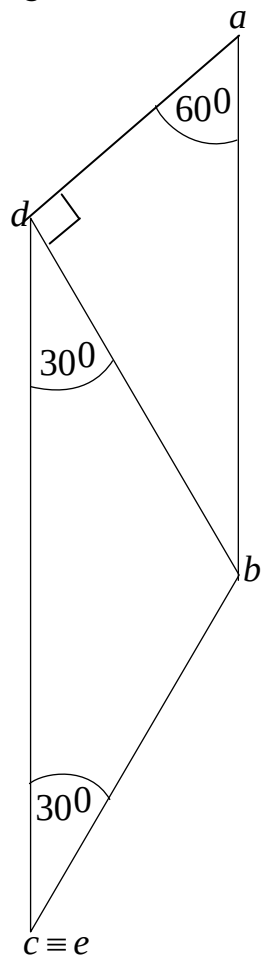
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75

(b)



ප්‍රධාන සටහන



දිශාව	ප්‍රත්‍යාබලය		
	නම	ආතති	තෙරපුම්
AB	ad	$W/2$	–
BC	bd	–	$\sqrt{3}W/2$
CD	be	–	$\sqrt{3}W/2$
DE	bc	–	$\sqrt{3}W/2$
DA	ce	–	–
CA	de	$3W/2$	–

බල



30

ආතති/තෙරපුම්



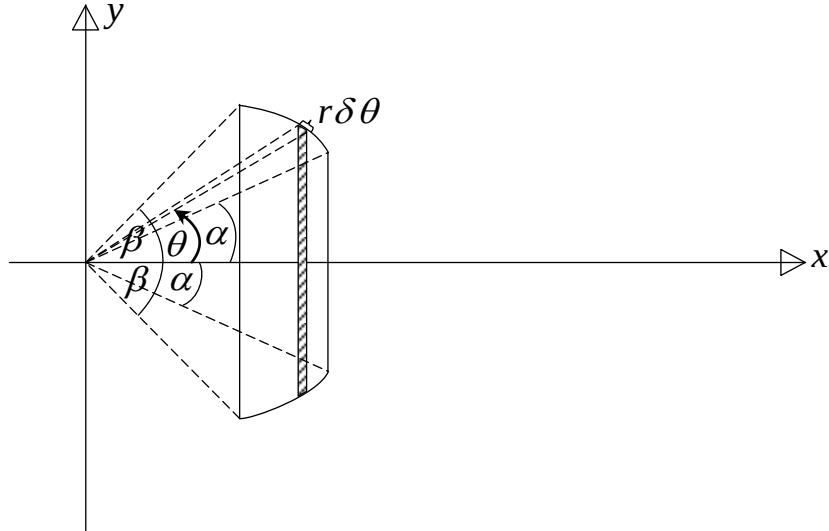
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16. (i)



$$\delta m = 2\pi r \sin \theta r \delta \theta \sigma$$

$$\text{ස්කන්ධය} = \int_{\alpha}^{\beta} 2\pi r^2 \sin \theta \cdot d\theta \sigma \quad (05)$$

$$= 2\pi r^2 \sigma \int_{\alpha}^{\beta} \sin \theta \cdot d\theta$$

$$= 2\pi r^2 \sigma [-\cos \theta]_{\alpha}^{\beta} \quad (05)$$

$$= 2\pi r^2 \sigma [-\cos \beta - (-\cos \alpha)]$$

$$\text{ස්කන්ධය} = 2\pi r^2 \sigma (\cos \alpha - \cos \beta) \quad (05)$$

මෙය x අක්ෂය වටා සමමිතික නිසා

ගුරුත්ව කේන්ද්‍රය x අක්ෂය මත පිහිටයි. (05)

$$\bar{x} = \frac{\int x dm}{\int dm}$$

$$= \frac{\int_{\alpha}^{\beta} r \cdot \cos \theta \cdot 2\pi r^2 \sin \theta \cdot d\theta \sigma}{\int_{\alpha}^{\beta} 2\pi r^2 \sin \theta d\theta \sigma} \quad (05)$$

$$= \frac{r \int_{\alpha}^{\beta} \sin \theta \cdot \cos \theta d\theta}{\int_{\alpha}^{\beta} \sin \theta \cdot d\theta}$$

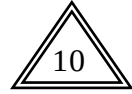
$$= \frac{r \left[\frac{\sin^2 \theta}{2} \right]_{\alpha}^{\beta}}{[-\cos \theta]_{\alpha}^{\beta}} \quad (05)$$

$$\bar{x} = \frac{r \left[\frac{\sin^2 \theta}{2} \right]_{\alpha}^{\beta}}{[-\cos \theta]_{\alpha}^{\beta}}$$

$$\begin{aligned}
&= \frac{r}{2} \left[\sin^2 \beta - \sin^2 \alpha \right] \\
&= \frac{r}{2} \left[-\cos \beta - (-\cos \alpha) \right] \\
&= \frac{r}{2} \left[(1 - \cos^2 \beta) - (1 - \cos^2 \alpha) \right] \\
&= \frac{r}{2} \left[\cos \alpha - \cos \beta \right] \\
\bar{x} &= \frac{\frac{r}{2} [\cos \alpha - \cos \beta] (\cos \alpha + \cos \beta)}{(\cos \alpha - \cos \beta)} \quad (05) \\
&= \underline{\underline{\frac{r}{2} (\cos \alpha + \cos \beta)}} \quad (05)
\end{aligned}$$



$$\begin{aligned}
m &= 2\pi r^2 \sigma (\cos \alpha - \cos \beta) \\
\text{අරය } r/2 \text{ නම්,} \\
2\pi \left(\frac{r}{2} \right)^2 \sigma (\cos \alpha - \cos \beta) &= \frac{m}{4} \text{ වේ.} \quad (05)
\end{aligned}$$

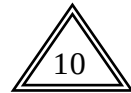


(ii) කුඩා වළල්ලේ ස්කන්ධ කේන්ද්‍රය

$$\begin{aligned}
&= \frac{r}{2} \cos \alpha - \frac{r/2}{2} [\cos \alpha + \cos \beta] \quad (05) \\
&= \frac{r}{2} \left[\cos \alpha - \frac{\cos \alpha}{2} - \frac{\cos \beta}{2} \right] \\
&= \frac{r}{2} \left[\frac{\cos(\pi/4)}{2} - \frac{\cos(\pi/3)}{2} \right]
\end{aligned}$$

වළල්ලේ ස්කන්ධ කේන්ද්‍රය

$$\begin{aligned}
&= \frac{r}{2} \left[\frac{1}{2\sqrt{2}} - \frac{1}{4} \right] \\
&= \frac{r}{2} \left[\frac{1}{\sqrt{2}} + \frac{1}{2\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{4} \right] \\
&= \frac{r}{2} \left[\frac{1}{\sqrt{2}} - \frac{(-2 + 4 + \sqrt{2})}{4\sqrt{2}} \right] \\
&= \frac{r}{2} \left[\frac{1}{\sqrt{2}} - \frac{2 + \sqrt{2}}{4\sqrt{2}} \right] \\
&= \underline{\underline{\frac{r}{2} \left[\frac{1}{\sqrt{2}} - k \right]}} \quad (05)
\end{aligned}$$



(iii) විශාල වළල්ලේ O සිට ස්කන්ධ කේන්ද්‍රය

$$= r \cos \alpha - \frac{r}{2} [\cos \alpha + \cos \beta] \quad (05)$$

$$= r \left[\frac{\cos \alpha}{2} - \frac{\cos \beta}{2} \right]$$

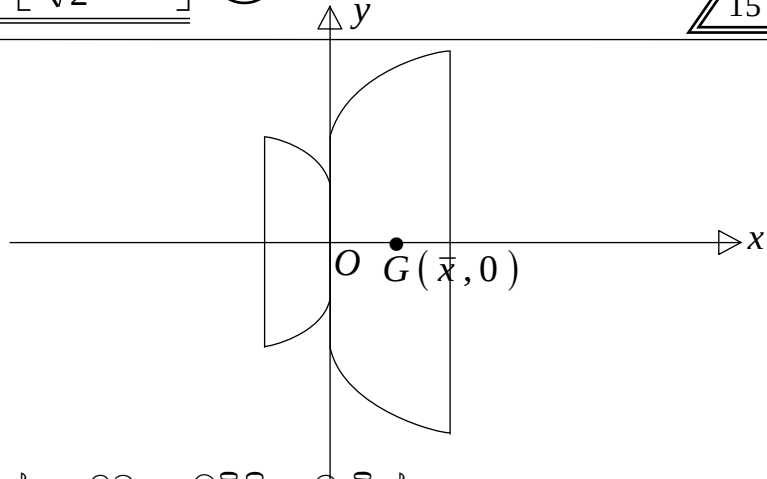
$$= r \left[\frac{\cos \pi/4}{2} - \frac{\cos (\pi/3)}{2} \right]$$

$$= r \left[\frac{1}{\sqrt{2}} + \frac{1}{2\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{4} \right] \quad (05)$$

$$= r \left[\frac{1}{\sqrt{2}} - \frac{2 + \sqrt{2}}{4\sqrt{2}} \right]$$

$$= r \left[\frac{1}{\sqrt{2}} - k \right] \quad (05)$$

15



මෙය x අක්ෂය වටා සමමිතික බැවින්
ගුරුත්ව කේන්ද්‍රය x අක්ෂය මත පිහිටයි. (05)

වස්තුව	m_i	x_i
	$m \quad (05)$	$r \left(\frac{1}{\sqrt{2}} - k \right) \quad (05)$
	$m/4 \quad (05)$	$-\frac{r}{2} \left(\frac{1}{\sqrt{2}} - k \right) \quad (05)$
	$m/4 \quad (05)$	$0 \quad (05)$
සංයුක්තය	$3m/2 \quad (05)$	$\bar{x}$

$$(\sum m_i) \bar{x} = (\sum m_i x_i)$$

$$\frac{3m}{2}(\bar{x}) = mr\left(\frac{1}{\sqrt{2}} - k\right) + \frac{m}{4}\left(-\frac{r}{2}\left(\frac{1}{\sqrt{2}} - k\right)\right) + \frac{m}{4}(0) \quad (10)$$

$$\div m \Rightarrow$$

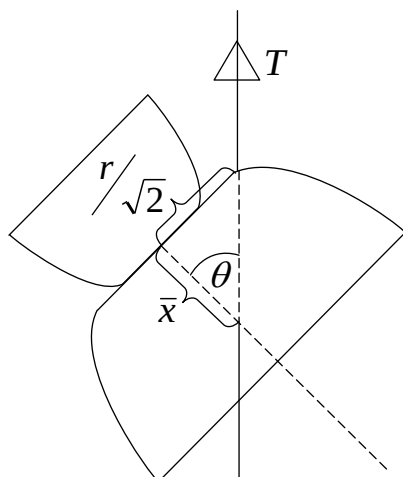
$$\frac{3\bar{x}}{2} = r\left(\frac{1}{\sqrt{2}} - k - \frac{1}{8\sqrt{2}} + \frac{k}{8}\right) \quad (05)$$

$$\bar{x} = \frac{2r}{3}\left(\frac{8-1}{8\sqrt{2}} - \frac{8k-k}{8}\right)$$

$$\bar{x} = \frac{r}{12}\left(\frac{7}{\sqrt{2}} - 7k\right)$$

$$\bar{x} = \frac{7r}{12}\left(\frac{1}{\sqrt{2}} - k\right) \quad (05)$$

60



$$\tan \theta = \frac{r/\sqrt{2}}{\bar{x}}$$

$$\tan \theta = \frac{r/\sqrt{2}}{7r/12\left(\frac{1}{\sqrt{2}} - k\right)} \quad (10)$$

$$\theta = \tan^{-1}\left(\frac{12}{7\sqrt{2}\left(\frac{1}{\sqrt{2}} - k\right)}\right) \quad (05)$$

150

15

150

17. (a)

$$A_1$$

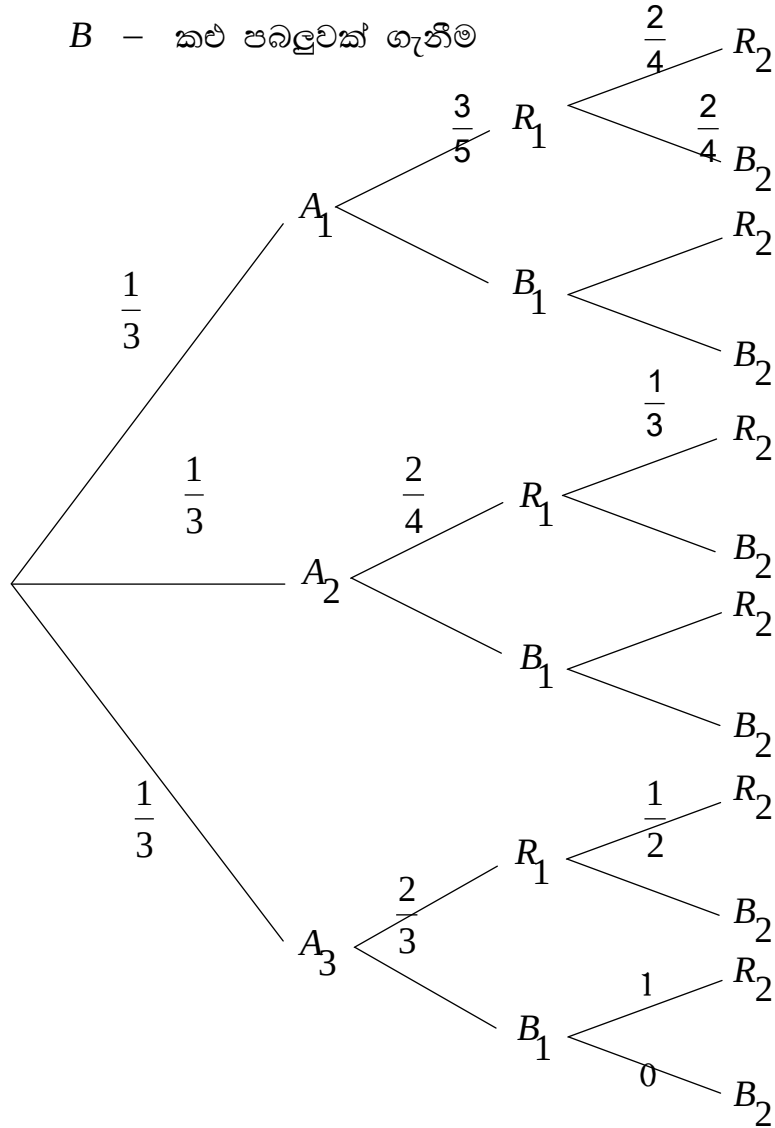
රතු	-	3
කළු	-	2

$$A_2$$

රතු	-	2
කළු	-	2

$$A_3$$

රතු	-	2
කළු	-	1

 R - රතු පබළුවක් ගැනීම B - කළු පබළුවක් ගැනීම

$$\begin{aligned}
 \text{(i)} \quad & P(R_1 \cap R_2) = P(R_1 \cap R_2 \cap A_1) + P(R_1 \cap R_2 \cap A_2) \\
 & \quad \quad \quad + P(R_1 \cap R_2 \cap A_3) \\
 & = P(R_2 | A_1 \cap R_1) \cdot P(A_1 \cap R_1) \\
 & \quad \quad \quad + P(R_2 | A_2 \cap R_1) \cdot P(A_2 \cap R_1) \\
 & \quad \quad \quad + P(R_2 | A_3 \cap R_1) \cdot P(A_3 \cap R_1) \\
 & = P(R_2 | A_1 \cap R_1) \cdot P(R_1 | A_1) \cdot P(A_1) \\
 & \quad \quad \quad + P(R_2 | A_2 \cap R_1) \cdot P(R_1 | A_2) \cdot P(A_2) \\
 & \quad \quad \quad + P(R_2 | A_3 \cap R_1) \cdot P(R_1 | A_3) \cdot P(A_3) \\
 & = \underbrace{\left(\frac{2}{4} \times \frac{3}{5} \times \frac{1}{3} \right)}_{(10)} + \underbrace{\left(\frac{1}{3} \times \frac{2}{4} \times \frac{1}{3} \right)}_{(10)} + \underbrace{\left(\frac{1}{2} \times \frac{2}{3} \times \frac{1}{3} \right)}_{(10)} \\
 & = \frac{1}{10} + \frac{1}{18} + \frac{1}{9} \\
 & = \frac{1}{10} + \frac{3}{18} \\
 & = \frac{1}{10} + \frac{1}{6} \\
 & = \frac{3+5}{30} \\
 & = \frac{8}{30} \\
 & = \frac{4}{15} \quad (10)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad P(A_3 | R_1 \cap R_2) &= \frac{P(A_3 \cap R_1 \cap R_2)}{P(R_1 \cap R_2)} \\
 &= \frac{P(R_2 | R_1 \cap A_3) \cdot P(R_1 \cap A_3)}{P(R_1 \cap R_2)} \\
 &= \frac{P(R_2 | R_1 \cap A_3) \cdot P(R_1 | A_3) \cdot P(A_3)}{P(R_1 \cap R_2)} \\
 &= \frac{\frac{1}{2} \times \frac{2}{3} \times \frac{1}{3}}{\frac{4}{15}} \\
 &= \frac{\frac{1}{9}}{\frac{4}{15}} \\
 &= \frac{5}{12}
 \end{aligned}$$

(05)

(05)

(05)

(05)

20
70

(b)

උස (1m වලින්)	f_i	x_i	y_i	$f_i y_i$	$f_i y_i^2$
5 – 13	2	9	-2	-4	8
13 – 21	f_1	17	-1	-5	5
21 – 29	9	25	0	0	0
29 – 37	f_2	33	1	3	3
37 – 45	1	41	2	2	4
				-4	20

$$y_i = \frac{x_i - a}{c} ; a = 25 \quad c = 8$$

15

මානය = 24.2 නිසා මාන පන්තිය 21 – 29 වේ.

$$M_O = L_1 + \frac{\Delta_1}{\Delta_1 + \Delta_2} c$$

$$24.2 = 21 + \frac{9 - f_1}{9 - f_1 + 9 - f_2} (8) \quad (10)$$

$$3.2(18 - f_1 - f_2) = (9 - f_1)8$$

$$4.8 f_1 - 3.2 f_2 = 14.4 \quad (05)$$

$$3 f_1 - 2 f_2 = 9 \quad (1)$$

මධ්‍යස්ථය = $\frac{71}{3}$ නිසා මධ්‍යස්ථ පන්තිය 21–29 වේ.

$$M_d = L_1 + \frac{c}{f m} \left(\frac{n}{2} - F_1 \right)$$

$$\frac{71}{3} = 21 + \frac{8}{9} \left(\frac{12 + f_1 + f_2}{2} - (2 + f_1) \right) \quad (10)$$

$$3 = \left(\frac{8 - f_1 + f_2}{2} \right)$$

$$f_1 - f_2 = 2 \quad (2) \quad (05)$$

$$(1) - (2) \times 2 \Rightarrow f_1 = 9 - 4$$

$$f_1 = 5 \quad (05)$$

$$f_2 = 3 \quad (05)$$



$$\bar{y} = \frac{\sum f_i y_i}{\sum f_i}$$

$$= \frac{-4}{20}$$

$$= -0.2$$

$$\bar{x} = a + c \bar{y}$$

$$\bar{x} = 25 + 8(-0.2) \quad (05)$$

$$= 23.4 \quad (05)$$

$$\begin{aligned}
 \sigma^2 &= |c|^2 \left(\frac{\sum f_i y_i^2}{\sum f_i} - \bar{y}^2 \right) \\
 &= 8^2 \left(\frac{20}{20} - \left(\frac{-4}{20} \right)^2 \right) \textcircled{10} \\
 &= 64(1 - 0.04) \\
 &= \underline{\underline{61.44}} \textcircled{05}
 \end{aligned}$$

